Laboratory Work 5 Activity | Applying DAX Fundamentals in Power BI: From Measures to Contextual Analysis

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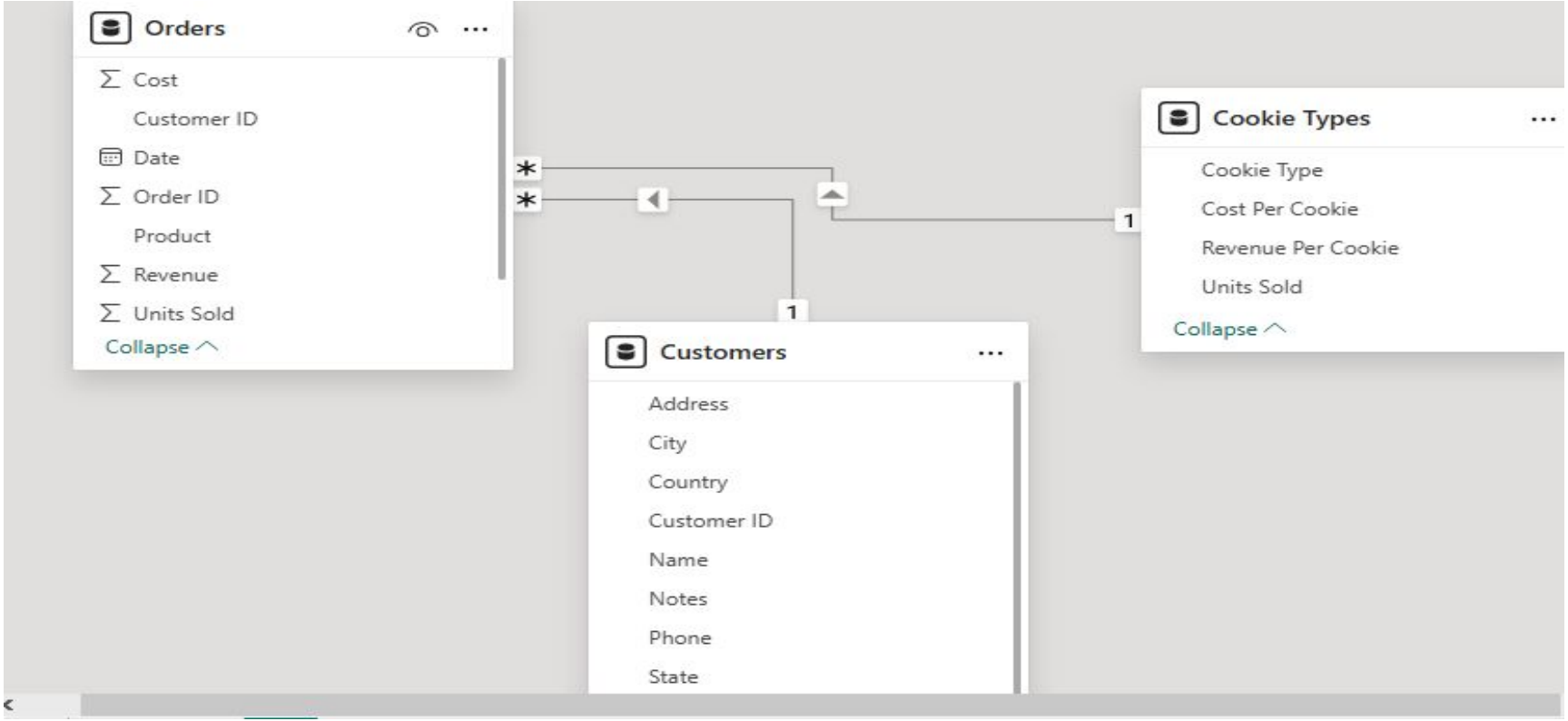
Instructor

Model View of Orders, Customers and Cookie Types



Formula 1: Total Revenue Measure

```
Total Revenue = SUM(Orders[Revenue])
```

Customer ID	Order ID	Product	Units Sold	Date	Revenue	Cost
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948
3	684759	Chocolate Chip	2518	Monday, June 1, 2020	12590	5036
3	183251	Chocolate Chip	1513	Tuesday, December 1, 2020	7565	3026
3	600124	Chocolate Chip	1030	Friday, May 1, 2020	5150	2060
3	562219	Chocolate Chip	1514	Saturday, February 1, 2020	7570	3028
3	387444	Chocolate Chip	787	Monday, June 1, 2020	3935	1574
3	365463	Chocolate Chip	1728	Friday, May 1, 2020	8640	3456
3	505159	Chocolate Chip	2145	Tuesday, October 1, 2019	10725	4290
3	300303	Chocolate Chip	1084	Tuesday, December 1, 2020	5420	2168
3	578401	Chocolate Chip	689	Monday, June 1, 2020	3445	1378
3	365552	Chocolate Chip	1570	Monday, June 1, 2020	7850	3140
3	239419	Chocolate Chip	4251	Wednesday, January 1, 2020	21255	8502
3	549329	Chocolate Chip	2918	Friday, May 1, 2020	14590	5836
3	288851	Chocolate Chip	2988	Wednesday, July 1, 2020	14940	5976
3	255145	Chocolate Chip	2074	Tuesday, September 1, 2020	10370	4148
3	538134	Chocolate Chip	1514	Tuesday, October 1, 2019	7570	3028
3	817134	Chocolate Chip	274	Tuesday, December 1, 2020	1370	548
3	697568	Chocolate Chip	1138	Tuesday, December 1, 2020	5690	2276
3	508782	Chocolate Chip	2992	Sunday, March 1, 2020	14960	5984
3	697895	Chocolate Chip	1228	Tuesday, October 1, 2019	6140	2456
3	691331	Chocolate Chip	1389	Tuesday, October 1, 2019	6945	2778
3	568366	Chocolate Chip	1496	Monday, June 1, 2020	7480	2992
3	176592	Chocolate Chip	2299	Tuesday, October 1, 2019	11495	4598

Orders (700 rows) Column: Total Revenue (0 distinct values)

Data

Search

- Cookie Types
- Customers
- Orders
  - Cost
  - Customer ID
  - Date
  - Order ID
  - Product
  - Revenue
  - Total Revenue
  - Units Sold

Formula 2: Total Cost Measure

```
1 Total Cost = SUM(Orders[Cost])
```

	Order ID	Product	Units Sold	Date	Revenue	Cost
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948
3	684759	Chocolate Chip	2518	Monday, June 1, 2020	12590	5036
3	183251	Chocolate Chip	1513	Tuesday, December 1, 2020	7565	3026
3	600124	Chocolate Chip	1030	Friday, May 1, 2020	5150	2060
3	562219	Chocolate Chip	1514	Saturday, February 1, 2020	7570	3028
3	387444	Chocolate Chip	787	Monday, June 1, 2020	3935	1574
3	365463	Chocolate Chip	1728	Friday, May 1, 2020	8640	3456
3	505159	Chocolate Chip	2145	Tuesday, October 1, 2019	10725	4290
3	300303	Chocolate Chip	1084	Tuesday, December 1, 2020	5420	2168
3	578401	Chocolate Chip	689	Monday, June 1, 2020	3445	1378
3	365552	Chocolate Chip	1570	Monday, June 1, 2020	7850	3140
3	239419	Chocolate Chip	4251	Wednesday, January 1, 2020	21255	8502
3	549329	Chocolate Chip	2918	Friday, May 1, 2020	14590	5836
3	288851	Chocolate Chip	2988	Wednesday, July 1, 2020	14940	5976
3	255145	Chocolate Chip	2074	Tuesday, September 1, 2020	10370	4148
3	538134	Chocolate Chip	1514	Tuesday, October 1, 2019	7570	3028
3	817134	Chocolate Chip	274	Tuesday, December 1, 2020	1370	548
3	697568	Chocolate Chip	1138	Tuesday, December 1, 2020	5690	2276
3	508782	Chocolate Chip	2992	Sunday, March 1, 2020	14960	5984
3	697895	Chocolate Chip	1228	Tuesday, October 1, 2019	6140	2456
3	691331	Chocolate Chip	1389	Tuesday, October 1, 2019	6945	2778
3	568366	Chocolate Chip	1496	Monday, June 1, 2020	7480	2992
3	176592	Chocolate Chip	2299	Tuesday, October 1, 2019	11495	4598

) Column: Total Cost (0 distinct values)

Data

Search

- Cookie Types
- Customers
- Orders
  - Cost
  - Customer ID
  - Date
  - Order ID
  - Product
  - Revenue
  - Total Cost
  - Total Revenue
  - Units Sold

Formula 3: Profit Measure

```
1 Profit = [Total Revenue] - [Total Cost]
```

	Order ID	Product	Units Sold	Date	Revenue	Cost
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948
3	684759	Chocolate Chip	2518	Monday, June 1, 2020	12590	5036
3	183251	Chocolate Chip	1513	Tuesday, December 1, 2020	7565	3026
3	600124	Chocolate Chip	1030	Friday, May 1, 2020	5150	2060
3	562219	Chocolate Chip	1514	Saturday, February 1, 2020	7570	3028
3	387444	Chocolate Chip	787	Monday, June 1, 2020	3935	1574
3	365463	Chocolate Chip	1728	Friday, May 1, 2020	8640	3456
3	505159	Chocolate Chip	2145	Tuesday, October 1, 2019	10725	4290
3	300303	Chocolate Chip	1084	Tuesday, December 1, 2020	5420	2168
3	578401	Chocolate Chip	689	Monday, June 1, 2020	3445	1378
3	365552	Chocolate Chip	1570	Monday, June 1, 2020	7850	3140
3	239419	Chocolate Chip	4251	Wednesday, January 1, 2020	21255	8502
3	549329	Chocolate Chip	2918	Friday, May 1, 2020	14590	5836
3	288851	Chocolate Chip	2988	Wednesday, July 1, 2020	14940	5976
3	255145	Chocolate Chip	2074	Tuesday, September 1, 2020	10370	4148
3	538134	Chocolate Chip	1514	Tuesday, October 1, 2019	7570	3028
3	817134	Chocolate Chip	274	Tuesday, December 1, 2020	1370	548
3	697568	Chocolate Chip	1138	Tuesday, December 1, 2020	5690	2276
3	508782	Chocolate Chip	2992	Sunday, March 1, 2020	14960	5984
3	697895	Chocolate Chip	1228	Tuesday, October 1, 2019	6140	2456
3	691331	Chocolate Chip	1389	Tuesday, October 1, 2019	6945	2778
3	568366	Chocolate Chip	1496	Monday, June 1, 2020	7480	2992
3	176592	Chocolate Chip	2299	Tuesday, October 1, 2019	11495	4598

s) Column: Profit (0 distinct values)

Data

Search

- Cookie Types
- Customers
- Orders
  - Cost
  - Customer ID
  - Date
  - Order ID
  - Product
  - Profit
  - Revenue
  - Total Cost
  - Total Revenue
  - Units Sold

Formula 4: Total Orders Measure

```
Total Orders = COUNT(Orders[Order ID])
```

	Order ID	Product	Units Sold	Date	Revenue	Cost
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948
3	684759	Chocolate Chip	2518	Monday, June 1, 2020	12590	5036
3	183251	Chocolate Chip	1513	Tuesday, December 1, 2020	7565	3026
3	600124	Chocolate Chip	1030	Friday, May 1, 2020	5150	2060
3	562219	Chocolate Chip	1514	Saturday, February 1, 2020	7570	3028
3	387444	Chocolate Chip	787	Monday, June 1, 2020	3935	1574
3	365463	Chocolate Chip	1728	Friday, May 1, 2020	8640	3456
3	505159	Chocolate Chip	2145	Tuesday, October 1, 2019	10725	4290
3	300303	Chocolate Chip	1084	Tuesday, December 1, 2020	5420	2168
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3	255145	Chocolate Chip	2074	Tuesday, September 1, 2020	10370	4148
3	538134	Chocolate Chip	1514	Tuesday, October 1, 2019	7570	3028
3	817134	Chocolate Chip	274	Tuesday, December 1, 2020	1370	548
3	697568	Chocolate Chip	1138	Tuesday, December 1, 2020	5690	2276
3	508782	Chocolate Chip	2992	Sunday, March 1, 2020	14960	5984
3	697895	Chocolate Chip	1228	Tuesday, October 1, 2019	6140	2456
3	691331	Chocolate Chip	1389	Tuesday, October 1, 2019	6945	2778
3	568366	Chocolate Chip	1496	Monday, June 1, 2020	7480	2992
3	176592	Chocolate Chip	2299	Tuesday, October 1, 2019	11495	4598

s) Column: Total Orders (0 distinct values)

Data

Search

- Cookie Types
- Customers
- Orders
  - Cost
  - Customer ID
  - Date
  - Order ID
  - Product
  - Profit
  - Revenue
  - Total Cost
  - Total Orders
  - Total Revenue
  - Units Sold

Formula 5: High Sales Measure

```
1 High Sales = CALCULATE([Total Revenue], Orders[Revenue] > 1000)
```

	Order ID	Product	Units Sold	Date	Revenue	Cost
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948
3	684759	Chocolate Chip	2518	Monday, June 1, 2020	12590	5036
3	183251	Chocolate Chip	1513	Tuesday, December 1, 2020	7565	3026
3	600124	Chocolate Chip	1030	Friday, May 1, 2020	5150	2060
3	562219	Chocolate Chip	1514	Saturday, February 1, 2020	7570	3028
3	387444	Chocolate Chip	787	Monday, June 1, 2020	3935	1574
3	365463	Chocolate Chip	1728	Friday, May 1, 2020	8640	3456
3	505159	Chocolate Chip	2145	Tuesday, October 1, 2019	10725	4290
3	300303	Chocolate Chip	1084	Tuesday, December 1, 2020	5420	2168
3	578401	Chocolate Chip	689	Monday, June 1, 2020	3445	1378
3	365552	Chocolate Chip	1570	Monday, June 1, 2020	7850	3140
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3	288851	Chocolate Chip	2988	Wednesday, July 1, 2020	14940	5976
3	255145	Chocolate Chip	2074	Tuesday, September 1, 2020	10370	4148
3	538134	Chocolate Chip	1514	Tuesday, October 1, 2019	7570	3028
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3	508782	Chocolate Chip	2992	Sunday, March 1, 2020	14960	5984
3	697895	Chocolate Chip	1228	Tuesday, October 1, 2019	6140	2456
3	691331	Chocolate Chip	1389	Tuesday, October 1, 2019	6945	2778
3	568366	Chocolate Chip	1496	Monday, June 1, 2020	7480	2992
3	176592	Chocolate Chip	2299	Tuesday, October 1, 2019	11495	4598

s) Column: High Sales (0 distinct values)

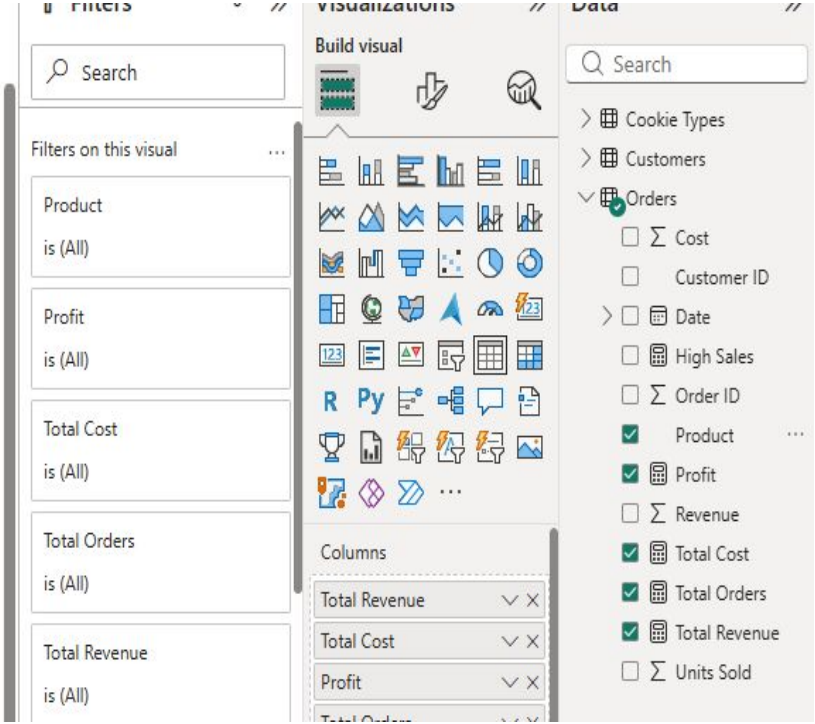
Data

Search

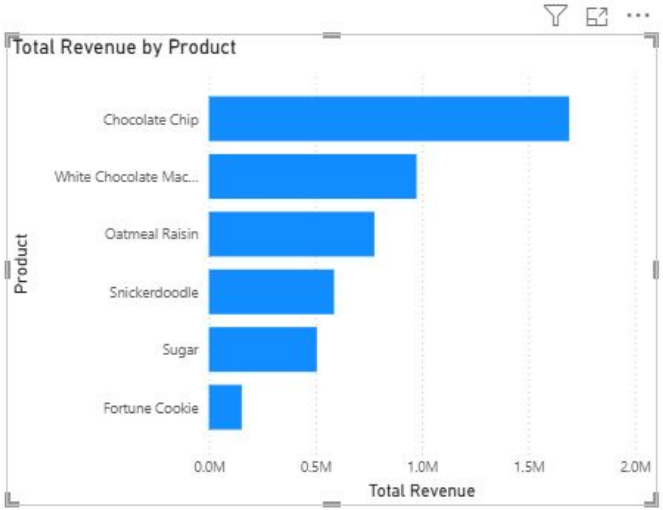
- Cookie Types
- Customers
- Orders
  - Cost
  - Customer ID
  - Date
  - High Sales
  - Order ID
  - Product
  - Profit
  - Revenue
  - Total Cost
  - Total Orders
  - Total Revenue
  - Units Sold

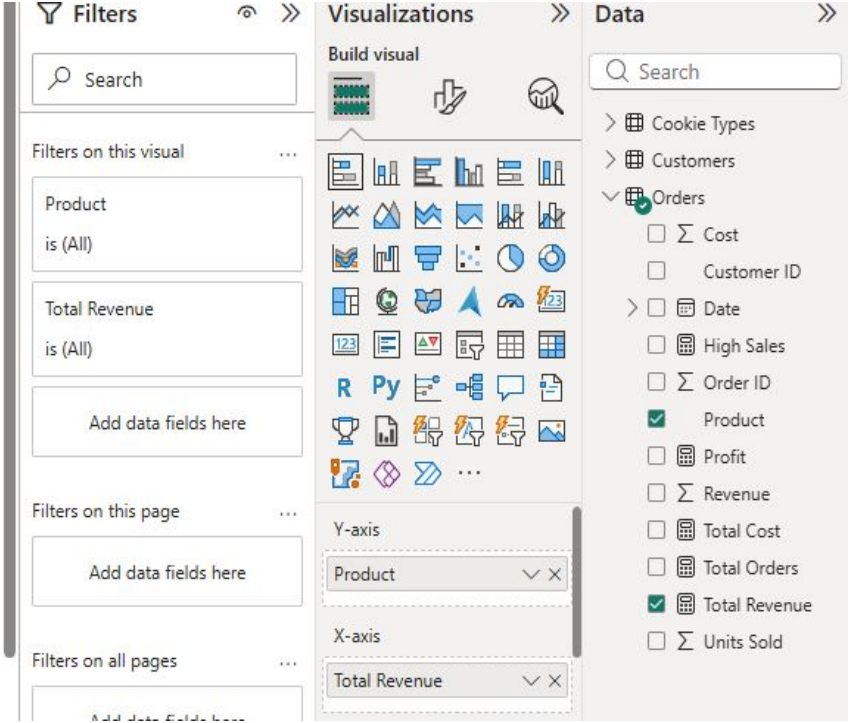
Visual 1: Summary Table

Total Revenue	Total Cost	Profit	Total Orders	Product
1,691,197.50	676479	1,014,718.50	202	Chocolate Chip
974,547.00	446667	527,880.00	109	White Chocolate Macadamia Nu
776,575.00	341694	434,881.00	94	Oatmeal Raisin
587,384.00	220266	367,118.00	93	Snickerdoodle
506,349.00	210976	295,373.00	109	Sugar
154,198.00	77098	77,100.00	93	Fortune Cookie
4,690,250.50	**1973180**	**2,717,070.50**	**700**	

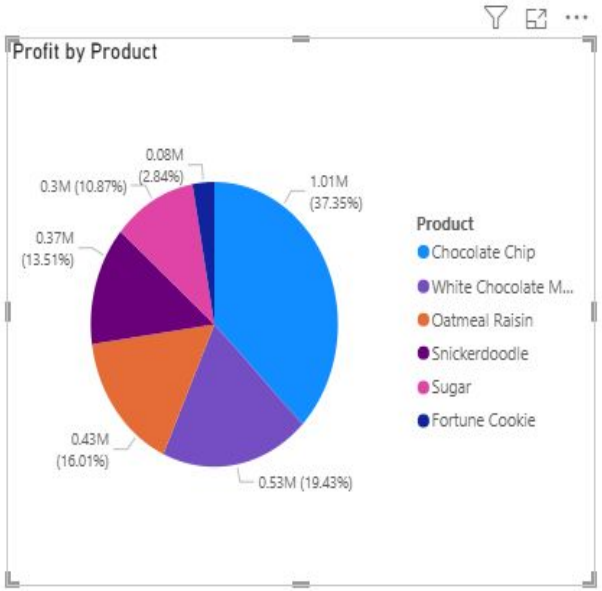


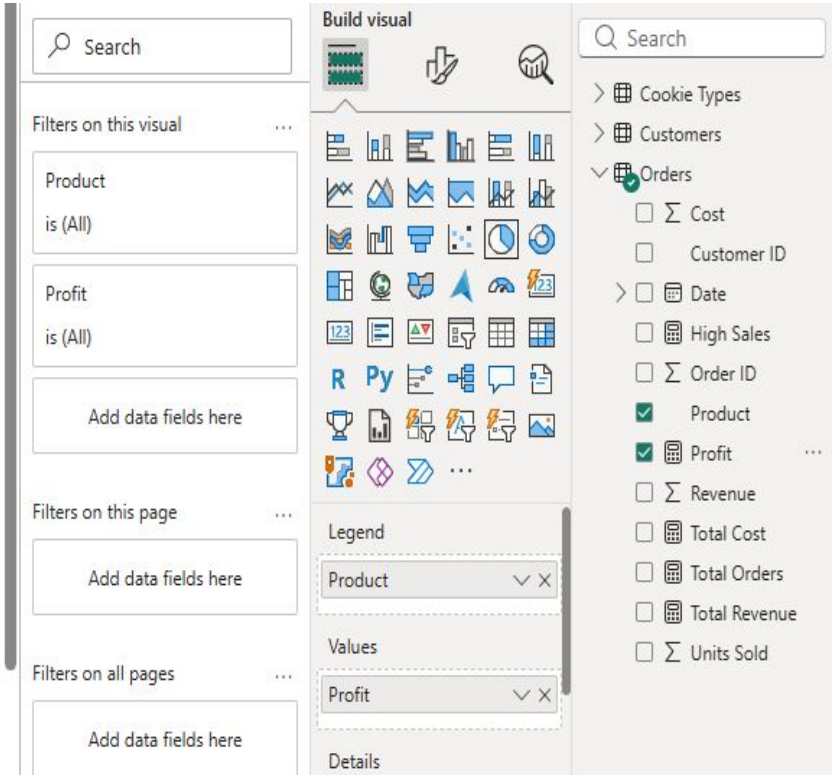
Visual 2: Total Revenue by Product Bar Chart



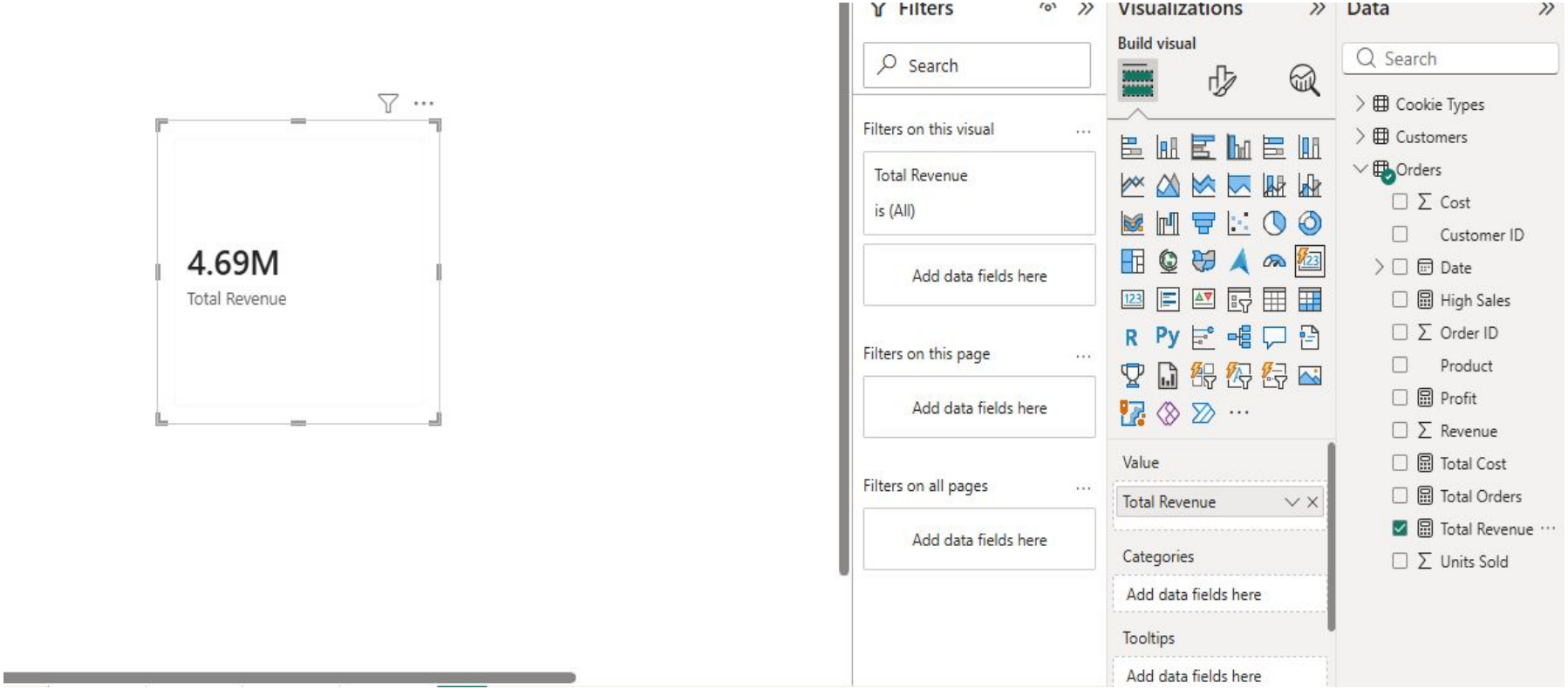


Visual 3: Profit by Product Pie Chart





Visual 4: Total Revenue Card



Guide Questions

1. Defining DAX: In your own words, how would you explain the difference between a standard Excel formula and a Data Analysis Expression (DAX) based on the tutorial?

A standard Excel formula works on a single cell or range in a flat sheet. DAX is different because it works across multiple related tables and adjusts its calculation depending on what filter or visual is active on the report.

2. The "Why" of DAX: Why can’t we just use the raw numbers from our imported dataset to create complex reports? What is the specific "gap" that DAX fills in Power BI?

Raw numbers alone cannot create dynamic calculations. DAX fills the gap by allowing us to create measures that automatically change based on filters, slicers, and visuals — something raw data cannot do on its own.

3. The Storage Difference: The video demonstrates creating both Calculated Columns and Measures. Which one actually increases the file size of your Power BI model, and why is this an important consideration for large datasets?

Calculated Columns increase the file size because they are stored inside the table for every single row. Measures do not increase file size because they are only calculated when a visual needs them.

4. Context Clues: * When you created a Calculated Column, the calculation happened row-by-row.
○ When you created a Measure, the result changed depending on which filters or visuals you clicked.
○ Question: Explain how "Filter Context" affects the result of a Measure in your final dashboard.

Filter Context means the result of a Measure changes depending on what is selected or filtered in the dashboard. For example, if you click on a specific product, the Total Revenue measure will only show the revenue for that product.

5. The RELATED Function: In the tutorial, the RELATED function is used. Why is this function necessary when working with multiple tables, and what must be established in the Model View for this function to work correctly?

The RELATED function is needed when data you need is in a different table. For it to work, the two tables must be connected by a relationship in the Model View using a common column like Product ID.

6. Deconstructing the Formula: Look at the following formula from the video:
Total Sales = SUM(Sales[Sales Amount]) Identify the Measure Name, the Function, the Table, and the Column. Why is it "best practice" to include the table name in the reference?

For `Total Sales = SUM(Sales[Sales Amount])`:

- **Measure Name** → Total Sales
- **Function** → SUM
- **Table** → Sales
- **Column** → Sales Amount
- Including the table name is best practice because it avoids confusion when multiple tables have columns with the same name.

7. Data Interpretation: Look at the visual you created using your DAX measures. If the "Total Profit" measure shows a negative number for a specific category, does that mean the DAX formula is "broken," or is it telling you something about the business data? How can you verify which it is?

A negative Total Profit does not mean the formula is broken — it means that category is losing money. You can verify it by checking if the Cost is higher than the Revenue for that category in your table visual.

8. Optimization: If you were tasked with calculating Profit Margin Percentage, would it be more efficient to create a Calculated Column for every row or a single Measure? Justify your answer based on what you learned about DAX efficiency.

It is more efficient to use a single Measure for Profit Margin Percentage. A Calculated Column would compute and store the value for every row, wasting memory. A Measure only calculates when needed, making the report faster and lighter.